

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0457

Eltopia 100 kV
May 02, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0457. The Interconnection Customer proposes a 115.5 MW solar photovoltaic facility interconnecting at the Eltopia 100 kV substation in the Indiana area. The request was received on April 06, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 266 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,075,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SEGS 5-HILLCREST 100.0 kV Line, the WAYNE TRACE-ALLEN 100.0 kV line loads to 132.8% of its 187 MVA rating. Project DFAX: 6.0%.

For the loss of BARRE-RAYMOND F. BARKER 100.0 kV, the SORENSON SUBSTATION-ELTOPIA 100.0 kV loads to 107.9% of its 166 MVA rating. Project DFAX: 100.0%.

For the loss of WAYNE TRACE SUBSTATION-DARLINGTON RD 100.0 kV, the HILLCREST-MULDOON MILL 100.0 kV loads to 103.4% of its 176 MVA rating. Project DFAX: 5.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WAYNE TRACE-ALLEN 100.0 kV	Rebuild line with higher-capacity conductor	\$4,399,000	12% (\$539,000)
SORENSON-ELTOPIA 100.0 kV L	Reconductor line	\$815,000	36% (\$295,000)
HILLCREST-MULDOON MILL 100.0 kV	Reconductor line	\$2,530,000	10% (\$241,000)
	Total allocated to project		\$1,075,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.